MIL-C-24246A(SH)
24 May 1976
SUPERSEDING
MIL-C-24246(SHIPS)
21 July 1967
(See 6.6)

MILITARY SPECIFICATION

COIL ASSEMBLY, COMPASS COMPENSATING (DEGAUSSING), TYPE "K-2"

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers type "K-2" compass compensating coils which are used to neutralize the magnetic field due to degaussing in the vicinity of a magnetic compass.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-S-92 - Screws, Machine; Slotted, or Cross-Recessed or Hexagon Head.
FF-W-92 - Washer, Flat (Plain).
QQ-B-613 - Brass, Leaded and Non-leaded; Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626 - Brass, Leaded and Non-leaded; Rod, Shaped, Forgings and Flat Products With Finished Edges (Bar and Strip).
QQ-B-650 - Brazing Alloys, Copper, Copper-Zinc, and Copper-Phosphorus.
QQ-B-654 - Brazing Alloys, Silver.
QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
TT-L-58 - Lacquer; Spraying, Clear and Pigmented, (General Interior Use).
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.

MILITARY

MIL-W-583 - Wire, Magnet, Electrical.
MIL-I-631 - Synthetic-Resin Composition, Nonrigid.
MIL-E-699 - Enamel, Deck, Gray, Exterior (Formula No. 20).
MIL-R-900 - Rubber Gasket Material, 45 Durometer Hardness for Moderately Low Temperature Service.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915/14 - Cable, Electrical, 600 Volts, Type MHOF.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-P-997 - Plastic Material, Laminated, Thermosetting, Electrical Insulation; Sheets, Glass Cloth Silicone Resin.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-P-15037 - Plastic Sheets, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-I-16923 - Insulating Compound, Electrical, Embedding.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-C-20079 - Cloth, Glass; Tape, Textile Glass; and Thread Glass.
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-681 - Identification Coding and Application of Hook Up and Lead Wire.

FSC 5950

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DRAWING

NAVSHIPS
803-4759447 - Compass Compensating Coil Type K-2.

(Copies of specifications, standards, Drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.4).

3.2 General requirements. The "K-2" compass compensating coils shall conform to MIL-E-917, and Drawing 803-4759447 except as otherwise specified herein. In case of conflict between information shown on the drawing and this specification, the specification shall govern.

3.3 Material. Materials of construction shall be nonmagnetic and entirely suited for the intended application of mounting in the weather aboard ship.

3.3.1 Soft solder. Soft solder shall conform to QQ-S-571.

3.3.2 Brazing solder. Brazing solders shall conform to QQ-B-650.

3.3.3 Lacquer. Lacquer shall conform to TT-L-58.

3.3.4 Plastics. Plastic materials shall conform to MIL-P-997 and MIL-P-15037.

3.3.5 Magnet wire. Magnet wire shall conform to MIL-W-583.

3.3.6 Paint. Paint shall conform to MIL-E-699.

3.3.7 Paint primer. Paint primer shall conform to TT-P-645.

3.3.8 Fibrous glass. Fibrous glass cloth, tape, and thread shall conform to MIL-C-20079.

3.3.9 Brass. Brass components shall conform to QQ-B-613 or QQ-B-626, as applicable.

3.3.10 Synthetic rubber gaskets. Synthetic rubber gasket material shall conform to MIL-R-900.

3.3.11 Silver brazing alloy. Silver base brazing alloy shall conform to QQ-B-654.

3.3.12 Machine screws. Machine screws shall conform to FF-S-92.

3.3.13 Brass washers. Brass washers shall conform to FF-W-92.

3.4 Components. The type "K-2" compass compensating (degaussing) coil assembly shall consist of five coils arranged in a rigid subassembly and enclosed in a watertight container, a connection box, and two cable leads.

3.5 Coil assembly characteristics. The assembly shall have the characteristics specified in table I.

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Table I - Coil assembly.

	Heeling Coil	NE/SW Coil	NW/SE Coil
No. coils	1	2	2
No. turns	4	4	4
Outer Wdg #1	60	150	150
Wdg #2	60	75	75
Wdg #3	60	75	75
Inner Wdg #4	60	75	75
Wire sizes (AWG) single Glass covered formvar	# 18 (0.0468)	# 20 (0.0375 in)	# 20 (0.0375 in)
Turns per layer			
Outer Wdg #1	30-2 layers	25-6 layers	25-6 layers
Wdg #2	30-2 layers	25-3 layers	25-3 layers
Wdg #3	30-2 layers	25-3 layers	25-3 layers
Inner Wdg #4	30-2 layers	25-3 layers	25-3 layers

3.6 Dimensions. The dimensions of the type "K-2" compass compensating (degaussing) coil assembly shall be as specified on Drawing 803-4759447.

3.7 Enclosure. The coil subassembly shall be enclosed in a #14 gauge brass case in accordance with the approximate configuration and limiting dimensions shown on Drawing 803-4759447.

3.7.1 Watertight integrity. The enclosure shall be capable of withstanding the watertight test specified in 4.5.3, and shall not be affected by salt water.

3.7.2 Soldered joints. Soft soldered joints shall be a minimum and where used, shall act as a sealing agent only, not as a strength member.

3.7.3 Color. The enclosure shall be painted with gray paint as specified in 3.3.6.

3.8 Connection box. The connection box shall contain 24 pressure type connectors racked in groups of six for connecting winding leads to cable leads. Connections from the box shall be made directly to the coils by pressure type connectors.

3.9 Nonmagnetic material. Nonmagnetic materials shall be used throughout this assembly. Nonmagentic is defined as having a magnetic permeability of 2.0 after fabrication.

3.10 Filling compound. The entire unit, including the terminal box, shall be filled with a filling or embedding compound conforming to MIL-I-16923 in such a manner as to fill all void spaces in the enclosure.

3.11 Insulating materials. All assemblies shall be insulated for operation as specified herein in a moisture laden atmosphere; every precaution shall be taken against possible failure of the insulation due to combination of heat and moisture. All insulated electrical windings shall be impregnated by the vacuum-pressure process to insure a thorough penetration and a complete filling of all interstices with a protective resin or clear, baking, insulating varnish as specified in MIL-I-24092. All insulating materials shall be class B insulation as defined in MIL-E-917.

3.12 Electrical connections. Two 20-foot lengths of type MHOF-14 cable in accordance with MIL-C-915/14 shall be used for cable leads. The two cables shall enter the enclosure through two stuffing tubes which are a part of the connection box. Connections from the two cable leads shall be made directly to pressure-type or other nonmagnetic connectors within the connection box. Spare conductors shall be dead-ended in the terminal box.

3.12.1 The cable entering the connection box on the left side shall contain conductors from the NE1, NE2, NE3, NE4, H1, and H2 windings. The right cable shall contain conductors from the NW1, NW2, NW3, NW4, H3, and H4 windings (see Drawing 803-4759447). Each cable lead free end conductor shall be identified with a reusable plastic tag with black figures on a white background. Corresponding numbered windings of diametrically opposite NE/SW coils shall be connected in series such that the magnetic field produced by the windings will be aiding when measured on the axis of the assembly between the coils. Opposite NW/SE windings shall similarly be connected in series aiding. Individual internal conductors shall be distinctly color coded in accordance with MIL-STD-681. Conductor ends shall be marked to identify associated windings through the use of sleeving. Sleeving shall be type F tubing, or equal, in accordance with MIL-I-631. Identification markings shall be permanently indented, not stamped, on the sleeve in a contrasting color.

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3.13 Electrical requirements.

3.13.1 Magnetic strength, polarity, and symmetry. The completed assembly shall pass the magnetic tests specified in 4.5.1.

3.13.2 Insulation resistance. The insulation resistance, when measured as specified in 4.5.2, shall be not less than 150 megohms (MΩ).

3.13.3 Dielectric strength. There shall be no evidence of insulation breakdown after being subjected to the dielectric strength test specified in 4.5.4.

3.14 Shock. The completed assembly shall meet the requirements of MIL-S-901 for grade A, class I equipments (see 4.5.5).

3.15 Vibration. The equipment shall be designed to meet the vibration requirements of MIL-STD-167-1, type I environmental vibration from four to 33 hertz (Hz) (see 4.5.6).

3.16 Identification plate data. The following information and instructions shall be etched on a brass identification plate in accordance with MIL-P-15024 and MIL-P-15024/5.

"Type "K-2" compass compensating coils"

"INSTALLATION

Fit the assembly over the binnacle with the top of the heeling coil on a level with the compass card and with uniform spacing between coil and binnacle. Shim flinders bar from binnacle as necessary to provide clearance for heeling coil. Bolt each intercardinal coil to the binnacle at one point. Make certain that arrows on heeling coil are aligned with binnacle lubber lines.

"WIRING

Connect leads labeled "NE" to control resistors labeled "NE/SW" and those labeled "NW" to control resistors labeled "NW/SE" and those labeled "H" to control resistors labeled "H".

"COMPENSATION

On any heading move slider on control resistor labeled "H" until dip needle at compass position remains level with D/G current on or off. Then place ship's heading on NE or SW by compass to regulate control resistros labeled "NE/SW" on NW or SE by compass to regulate resistors labeled "NW/SE". Energize degaussing coil and move slider on corresponding control resistor until compass returns to initial reading. Repeat compensation for fine control.

"DO NOT EXCEED 1.4 amperes per winding. If flinders bar is aft, and coil assembly is rotated 180 degrees, wiring and compensation procedure are identical.

"CONTRACT NO._____

"NATIONAL STOCK NO._____

"SERIAL NO._____

"MFR._____"

3.16.1 The identification plate shall be attached on the cover of the connection box.

3.17 Technical data. The supplier shall prepare technical data in accordance with the ordering documents included in the contract or order (see 6.2.2) and as specified in 4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination of 4.4.1 and the tests specified in 4.5.

4.4 Quality conformance inspection.

4.4.1 Examination. Each unit coil assembly shall be subjected to surface examination to determine conformance with this specification. Surface examination shall include visual examination for defects and finish, workmanship, and measurements for external dimensional limitations specified on Drawing 803-4759447. Any item which fails to conform to the requirements shall be rejected.

4.4.2 Quality conformance tests. Sample coils shall be subjected to the tests of 4.5 with the exception of shock and vibration (4.5.5 and 4.5.6).

4.4.3 Sampling. As coil assemblies are produced and the number of finished units increases, sample coil assemblies shall be selected for quality conformance tests, as shown in table II. Until 12 have passed all of the tests, no failures are allowed. If any failure occurs, the cause must be found and a design change to correct the cause in all units produced shall be accomplished. Sampling shall begin again at the first line of the table and progress downward. At no time shall the cumulated number of failures exceed the number allowed for the number of units produced and the number of units tested as shown in the table. If the cumulated number of failures equals the disapproval number, all test data shall be reported to the command or agency concerned for review.

Table II - Sampling for quality conformance tests.

Cumulated production coil assemblies	Cumulating sample to be tested	Approval number	Disapproval number
Up to 5	3	0	1
6 to 9	4		
10 to 14	5		
15 to 20	6		
21 to 28	7		
29 to 36	8		
37 to 45	10		
46 to 56	11		
57 to 68	12		
69 to 81	13		
82 to 95	14		
96 to 110	15		
111 to 126	16	1	2
127 to 144	17		
145 to 163	19		
164 to 183	20		
184 to 203	21		
204 to 225	22	2	3
226 to 248	23		
249 to 273	24		
274 to 298	25		

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4.5 Test procedures.

4.5.1 Magnetic test. The completed coil assemblies shall be tested for minimum magnetic field intensity and for orthogonality of the axes of the coils. For the tests the four windings of the heeling coil shall be corrected in series aiding. The NE/SW and NW/SE coils shall be similarly connected. All magnetic field intensity measurements shall be made at the vertical center of the heeling coil in three mutually perpendicular directions:

(a) Along the axis of the heeling coil.
(b) Parallel to the axis of the NE/SW coils and in the plane at the center of the heeling coil.
(c) Parallel to the axis of the NW/SE coils and in the plane at the center of the heeling coil.

The magnetometer for making these measurements shall be oriented on the axis of and in the plane at the center of the heeling coil so as to cause its horizontal plane to be parallel to the plane of the heeling coil within 6 minutes of arc. The tests specified hereinafter shall be performed.

4.5.1.1 The heeling coil only shall be energized with 1.0 ampere, d.c. The intensity of the resultant magnetic field measured along the axis of the heeling coil shall be not less than 7570 millioersteds, but no more than 65 millioersteds shall be produced along the axes of the NE/SW and NW/SE coils.

4.5.1.2 The NE/SW coils only shall be energized with 1.0 ampere, d.c. The intensity of the resultant magnetic field measured along the axis of the NE/SW coils shall be not less than 940 millioersteds but no more than 8 millioersteds shall be produced along the axes of the NW/SE and the heeling coils.

4.5.1.3 The NW/SE coils only shall be energized with 1.0 ampere, d.c. The intensity of the resultant magnetic field measured along the axis of the NW/SE coils shall be not less than 940 millioersteds, but no more than 8 millioersteds shall be produced along the axes of the NE/SW and the heeling coils.

4.5.2 Insulation resistance. The insulation resistance of all twelve circuits shall be checked between each other and between each circuit and the case using a 600-volt megger or insulation tester. Coils with readings less than 150 MΩ will be rejected. This test shall be performed prior to and after completion of the shock and vibration test.

4.5.3 Watertight integrity. Each coil assembly shall be tested for watertight integrity of the enclosure, using external water pressure applied in a container fitted with stuffing tubes for entrance of cable leads. Insulation resistance measured between any two conductors of the cable lead shall be at least MΩ after the unit has been subjected to a pressure of 25 pound per square inch (lb/in^2) for 5 minutes. Soapy water shall be employed for this test. No water shall have penetrated the enclosure during this period.

4.5.4 Dielectric strength. The coil assembly shall be submitted to a dielectric test voltage of 1500 volts between windings or between any winding and the case for a period of 1 minute.

4.5.5 Shock. The completed assembly shall withstand the type A shock test of MIL-S-901 for grade A, class I equipments. Shock tests shall be conducted on a standard Navy shock testing machine for lightweight, medium weight, or heavyweight equipment in accordance with MIL-S-901 (see 3.14). Failure, defined as inability to perform principal functions shall result in rejection. Failure to perform principal functions is defined as follows:

(a) Breakage of any parts, including mounting bolts. Minor chipping or distortion of parts will be accepted when such chipping or distortion cannot in any way impair the operation or performance requirements of the equipment.
(b) Appreciable distrotion of any parts including the case and mounting appendages.
(c) Insulation resistance values lower than those permitted by this specification.
(d) Failure to pass the dielectric strength test.
(e) Failure to pass the magnetic tests required by this specification.

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4.5.6 Vibration. The equipment shall be tested in accordance with MIL-STD-167-1, type I environmental vibration from four to 33 Hz. The unit shall be tested under normal loaded conditions to duplicate shipboard installation. Submission of a detailed test "setup" for Navy review shall be required prior to any testing. Test reports shall be submitted to the Naval Ship Engineering Center, Ship Safety, Noise/Vibration Damage Control Section and Magnetic Defense Section.

4.6 Test reports. The supplier shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2). Results of all tests and detailed descriptions of any damage or malfunctions incurred, at what stage of the tests it occurred, as well as details of any corrective action, shall be furnished. The test report shall be a complete chronological record of all tests performed, and shall include all test data developed from electrical, magnetic; and shock and vibration tests. A copy of the first article test report shall also be furnished to the command or agency concerned.

4.7 Inspection of preparation for delivery. The inspection of the preservation-packaging, packing and marking shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2 (see 6.5).

5.1 Preservation-packaging, packing, and marking. The equipment, repair parts, and technical publications shall be preserved-packaged level A or C, packed level A, B, or C and marked in accordance with MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. This coil assembly is for use on U.S. Navy Ships to produce a magnetic field at the magnetic compass which is equal and opposite to that produced at the same position by the degaussing coils.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) The required delivery dates for drawings, parts, and equipment.
(c) When test reports are required (see 4.6).
(d) Levels of preservation-packaging and packing required (see 5.1).

6.2.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

Specification paragraph	Data requirements	Service	Applicable DID	Options
4.6	First Article Test Report	SH	UDI-T-23790	-------

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.3 Government furnished equipment. Procurement documents should specify Government furnished equipment when furnished by the Government. Tooling is available at the Ships Parts Control Center as number T691-9223 for the type K-2 compass compensating coil.

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6.4 First article inspection.

6.4.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 5950-N166)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24246A(SH)	COIL ASSEMBLY, COMPASS COMPENSATING (DEGAUSSING), TYPE "K-2"

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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NOTICE
OF REINSTATEMENT

INCH-POUND
MIL-C-24246A(SH)
NOTICE 2
6 October 1989
SUPERSEDING
NOTICE 1
2 November 1988

MILITARY SPECIFICATION

COIL ASSEMBLY, COMPASS COMPENSATING (DEGAUSSING), TYPE "K-2"

MIL-C-24246A(SH), dated 24 May 1976, is hereby reinstated and may be used for acquisition The Naval Sea Systems Command (SH) hereby assumes preparing activity responsibility in lieu of the Naval Electronic Systems Command (EC).

(Copies of the referenced federal and military specifications, standards, and handbooks are available from the Department of Defense Single Stock Point, Naval Publications and Forms Center (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Preparing activity:
Navy - SH
(Project 5950-N224-02)

AMSC N/A FSC 5950
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